NLP tasks

Task 1: Sentiment Analysis on product reviews

Description:

- Dataset (recommended): IMDb reviews or Amazon product reviews (Kaggle)
- Analyze product reviews to determine whether the sentiment is positive or negative
- Clean and preprocess text (e.g., lowercasing, remove stop words)
- Lemmatization (prefer over stemming)
- Convert text to numerical format using TF-IDF or count Vectorizer
- Train a binary classifier (e.g., logistic regression) and evaluate its performance

Handle:

- Emojis 😊 → convert to sentiment tokens
- Negations ("not good" → important!)
- HTML tags / URLs

Remove duplicates and very short reviews

Tools & libraries:

Python – Pandas _ NLTK or spacy – scikit-learn

Covered topics

Text classification | Sentiment analysis

Advanced:

- Word embeddings:
  - Word2Vec / GloVe
- Sentence embeddings:
  - BERT (optional stretch)

Model Building

Train multiple models and compare:

Baseline Models:

- Logistic Regression
- Naïve Bayes (MultinomialNB)
- Support Vector Machine (SVM)

Advanced Models:

- Random Forest / Gradient Boosting
- (Bonus) Fine-tune a transformer like BERT

Model Evaluation (More Rigorous)

Don’t stop at accuracy:

- Confusion Matrix
- Precision, Recall, F1-score
- ROC-AUC curve

Visualization (Bonus++)

Instead of basic plots:

- Word clouds for positive vs negative reviews
- Bar charts of most important features
- ROC curve visualization
- Confusion matrix heatmap

Model Comparison Report

Create a structured report:

Model	Accuracy	F1 Score	ROC-AUC	Notes
Logistic Regression				
Naïve Bayes				
SVM				

Insight requirement:

- Explain *why* one model performs better
- Discuss trade-offs (speed vs accuracy)

Step 9: Deployment Simulation (Advanced Bonus)

- Save model using joblib or pickle
- Build a simple function:

predict review ("This product is amazing!")

Optional:

- Wrap into a simple API using Flask or FastAPI